

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

6 Computation Theory (amp12)

- Lambda calculus (a) Explain why the Church-Rosser Theorem implies that any λ -term that is β -convertible ($=_\beta$) to a term in β -normal form is in fact β -reducible ($\rightarrow$) to one in β -normal form. [2 marks]

Answer: It is a corollary of the Church-Rosser Theorem that $M =_\beta N$ holds iff there exists N' with $M \rightarrow N'$ and $N \rightarrow N'$. Furthermore, if N is in β -normal form, then $N \rightarrow N'$ can only hold if N' is α -equivalent to N and hence is also in β -normal form. So given M , $M =_\beta N$ for some β -normal form N iff $M \rightarrow N'$ for some β -normal form N' .

- (b) Let Bnf denote the set of λ -terms that have a β -normal form. Give with justification an example of two closed λ -terms I and Ω with $I \in Bnf$ and $\Omega \notin Bnf$. [3 marks]

Answer: $I = \lambda x.x$ is closed and in β -normal form and hence $I \in Bnf$.

$\Omega = (\lambda x.xx)(\lambda x.xx)$ is closed and satisfies that $\Omega \rightarrow M$ holds iff $M = \Omega$; in particular Ω is not in β -normal form. It follows that $\Omega \rightarrow M$ holds iff $M = \Omega$ and hence from part (a) that $\Omega \notin Bnf$.

- (c) Suppose that $\#$ is a bijection between the set of all λ -terms and the set $\mathbb{N}$ of all natural numbers and that there are recursive functions $\alpha : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ and $\nu : \mathbb{N} \rightarrow \mathbb{N}$ satisfying for all λ -terms M, N and numbers n that

$$\alpha(\#(M), \#(N)) = \#(MN) \quad (1)$$

$$\nu(n) = \#(\underline{n}) \quad (2)$$

where the λ -term $\underline{n}$ is the n^{th} Church numeral. Writing $\ulcorner M \urcorner$ for $\#(M)$, show that there are closed λ -terms **App** and **Num** satisfying for all λ -terms M, N that

$$\mathbf{App} \ulcorner M \urcorner \ulcorner N \urcorner =_\beta \ulcorner MN \urcorner \quad (3)$$

$$\mathbf{Num} \ulcorner N \urcorner =_\beta \ulcorner \underline{\nu(N)} \urcorner \quad (4)$$

(Any general properties of partial recursive functions with respect to λ -calculus you use should be carefully stated, but need not be proved.) [6 marks]

Answer: We use the fact that the collection of partial recursive functions coincides with the collection of λ -definable functions. Hence there are closed λ terms **App** and **Num** representing α and ν . Hence for all $m, n \in \mathbb{N}$

$$\mathbf{App} \underline{m} \underline{n} =_\beta \underline{\alpha(m, n)} \quad (5)$$

$$\mathbf{Num} \underline{n} =_\beta \underline{\nu(n)} \quad (6)$$

and therefore for all λ -terms M and N

$$\mathbf{App} \ulcorner M \urcorner \ulcorner N \urcorner \triangleq \mathbf{App} \underline{\#(M)} \underline{\#(N)} \stackrel{(5)}{=}_\beta \underline{\alpha(\#(M), \#(N))} \stackrel{(1)}{=} \underline{\#(MN)} \triangleq \ulcorner MN \urcorner$$

$$\mathbf{Num} \ulcorner N \urcorner \triangleq \mathbf{Num} \underline{\#(N)} \stackrel{(6)}{=}_\beta \underline{\nu(\#(N))} \stackrel{(2)}{=} \underline{\#(\nu(N))} \triangleq \ulcorner \underline{\nu(N)} \urcorner$$

- (d) Consider the following property of a closed λ -term F (where $\ulcorner M \urcorner$ is as in part (c)):

$$\text{for all } \lambda\text{-terms } M, \quad \begin{cases} F \ulcorner M \urcorner =_{\beta} \underline{0} & \text{if } M \in Bnf \\ F \ulcorner M \urcorner =_{\beta} \underline{1} & \text{if } M \notin Bnf \end{cases} \quad (7)$$

Let $H = \lambda h. P \Omega I (F (\text{App } h (\text{Num } h)))$ where $P = \lambda x y f. f (\lambda z. y) x$ and Ω and I are as in part (b). By considering whether or not $H \ulcorner H \urcorner$ is in Bnf , deduce that there can be no λ -term F satisfying (7). [6 marks]

Answer: Suppose F satisfies (7). Note that

$$\begin{aligned} H \ulcorner H \urcorner &=_{\beta} P \Omega I (F (\text{App } \ulcorner H \urcorner (\text{Num } \ulcorner H \urcorner))) \\ &=_{\beta} P \Omega I (F (\text{App } \ulcorner H \urcorner \ulcorner H \urcorner)) && \text{by (4)} \\ &=_{\beta} P \Omega I (F \ulcorner H \urcorner H \urcorner) && \text{by (3)} \\ &=_{\beta} F \ulcorner H \urcorner H \urcorner (\lambda z. I) \Omega && \text{by definition of } P \end{aligned}$$

So if $H \ulcorner H \urcorner \in Bnf$, then by (7) $F \ulcorner H \urcorner H \urcorner =_{\beta} \underline{0}$ and hence from above and by definition of $\underline{0}$, $H \ulcorner H \urcorner =_{\beta} F \ulcorner H \urcorner H \urcorner (\lambda z. I) \Omega =_{\beta} \underline{0} (\lambda z. I) \Omega =_{\beta} \Omega$. Thus $H \ulcorner H \urcorner \in Bnf$ and $H \ulcorner H \urcorner =_{\beta} \Omega$; but Bnf is closed under $=_{\beta}$, so $\Omega \in Bnf$, contradicting the choice of Ω in part (b).

On the other hand, if $H \ulcorner H \urcorner \notin Bnf$, then by (7) $F \ulcorner H \urcorner H \urcorner =_{\beta} \underline{1}$ and hence from above and by definition of $\underline{1}$, $H \ulcorner H \urcorner =_{\beta} F \ulcorner H \urcorner H \urcorner (\lambda z. I) \Omega =_{\beta} \underline{1} (\lambda z. I) \Omega =_{\beta} (\lambda z. I) \Omega =_{\beta} I$. Thus $H \ulcorner H \urcorner =_{\beta} I \in Bnf$ and Bnf is closed under $=_{\beta}$, so $H \ulcorner H \urcorner \in Bnf$, contradiction.

So in either case we obtain a contradiction and so no such F can exist.

- (e) Deduce from part (d) that $\{\#(M) \mid M \in Bnf\}$ is an undecidable set of numbers. [3 marks]

Answer: Note that since $\#$ is a bijection, for all N we have $\#(N) \notin \{\#(M) \mid M \in Bnf\}$ when $N \notin Bnf$ (as well as $\#(N) \in \{\#(M) \mid M \in Bnf\}$ when $N \in Bnf$).

So if $\{\#(M) \mid M \in Bnf\}$ were decidable there would be a recursive function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ satisfying for all M that $\begin{cases} \varphi(\#(M)) = 0 & \text{if } M \in Bnf \\ \varphi(\#(M)) = 1 & \text{if } M \notin Bnf \end{cases}$; but then since recursive functions are λ -definable, φ would be represented by some closed λ -term F satisfying (7), which is impossible. So $\{\#(M) \mid M \in Bnf\}$ is undecidable.